

Learning semiconductor process logic

A from-scratch, step-token GPT predicts next steps and completions; a rule validator handles anomalies with explicit process-rule attribution.

0.722

OOD Top-1

train 12 / test 3

0.999

OOD Top-5

family-wise avg

0.263

Task 2 NED

lower is better

1.00

Task 3 F1

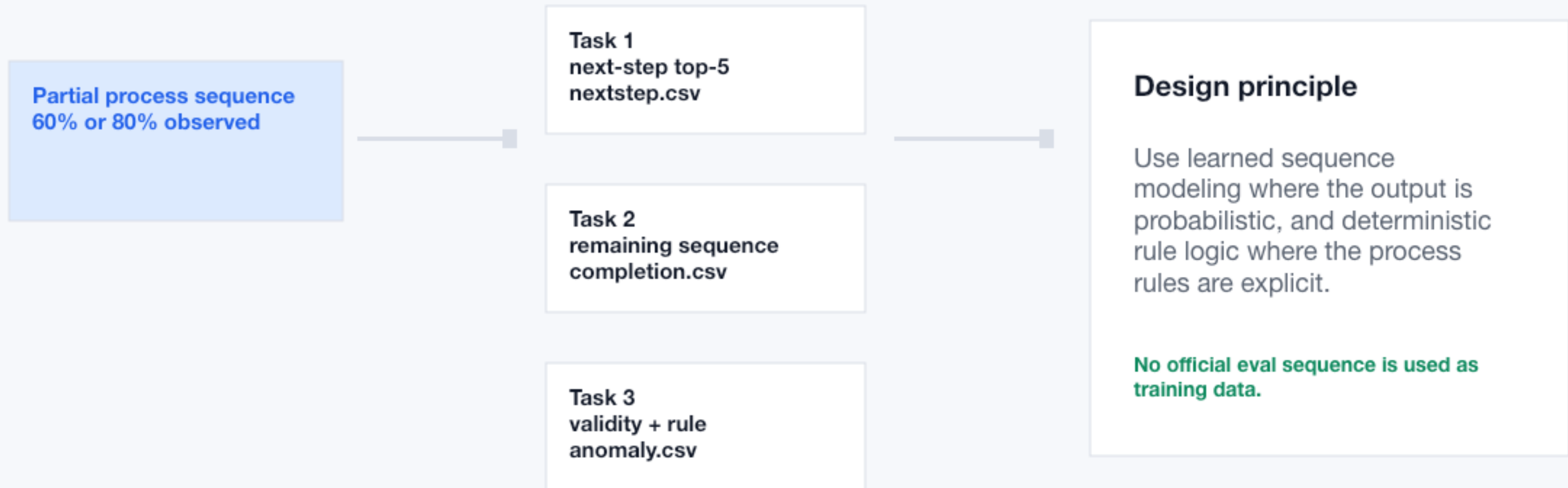
rule validator

Final package

- nextstep.csv
- completion.csv
- anomaly.csv
- trained checkpoint
- score reports

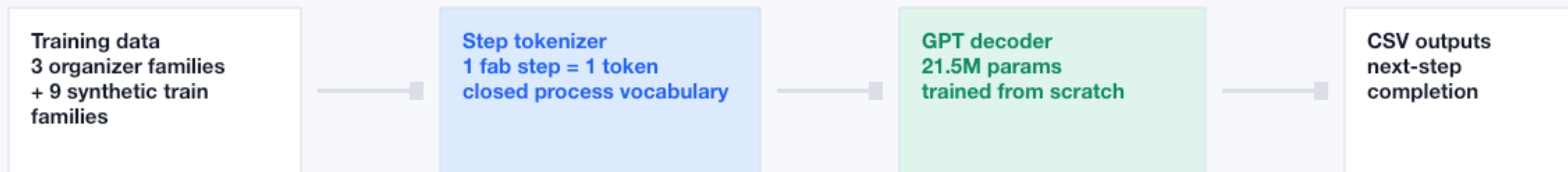
The challenge is three outputs from one process prefix

The jury scores files, but the engineering problem is process grammar generalization.



What we built: a small process model, not a generic LLM wrapper

Each process step is one token, so training optimizes exactly the scored unit.

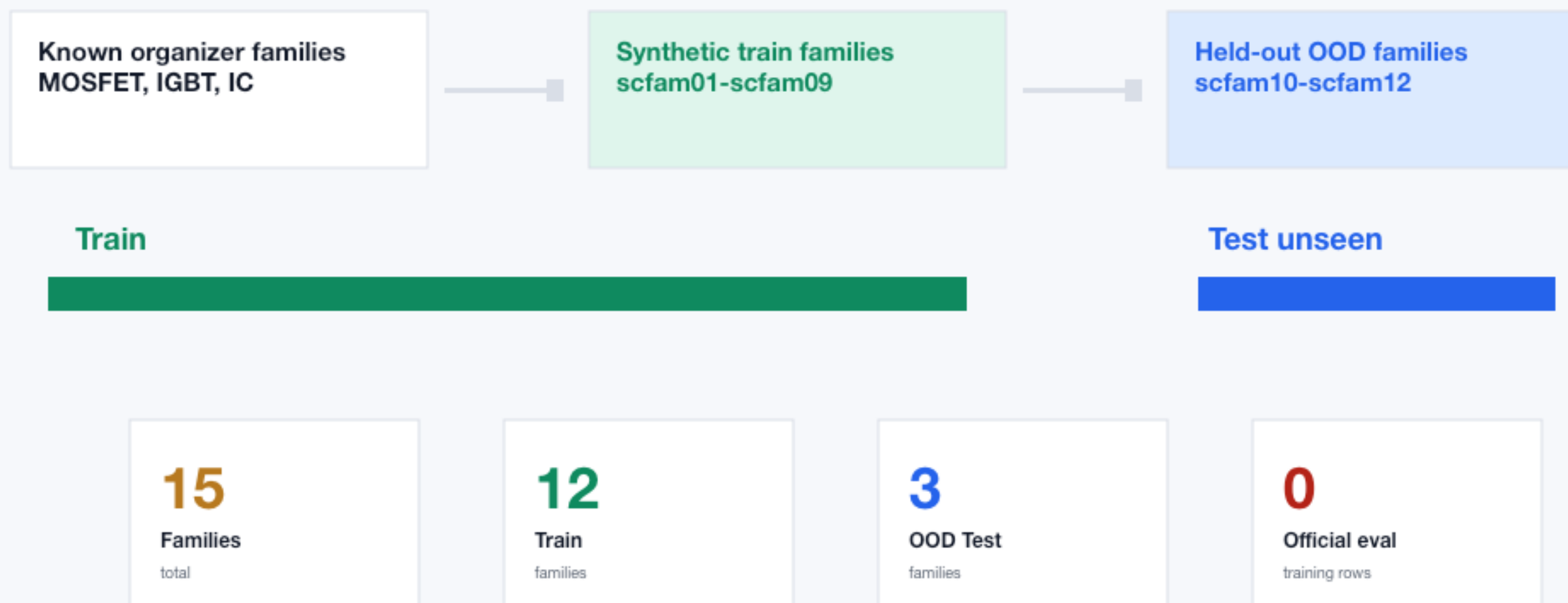


Anomaly path: organizer process rules -> deterministic validator -> validity, score, and official rule name

The n-gram baseline remains in the repo as a dependency-light fallback and an honest comparator.

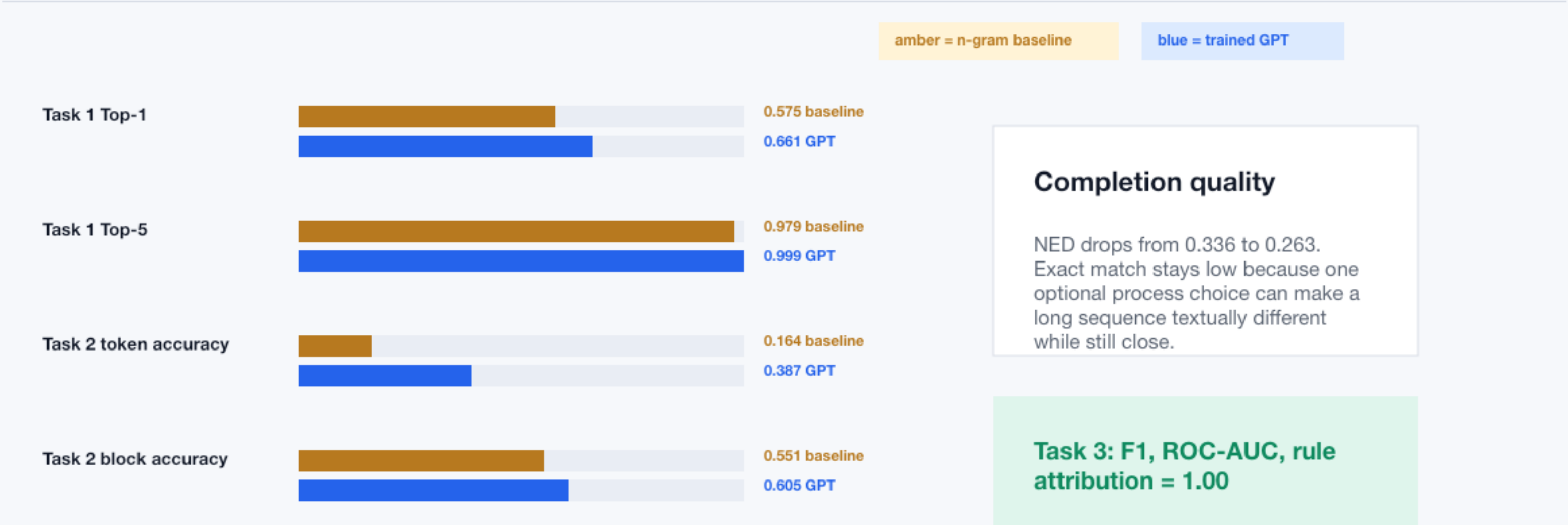
We evaluate for hidden-family behavior, not just self-eval fit

The final decision protocol is train on 12 families and report the average over 3 unseen families.



The trained model improves the scored tasks

Organizer scorer on our self-eval split, plus the frozen 15-family OOD benchmark.



Same prefix, different model behavior

A real self-eval MOSFET prefix at the 80% cut.

Prefix tail: DEPOSIT PASSIVATION | CURE PASSIVATION | MEASURE PASSIVATION QUALITY | OPEN PAD WINDOW | OPEN PAD WINDOW LITHO | DEVELOP PHOTORESIST | PASSIVATION ETCH PAD OPENING

Ground truth next step
STRIP PHOTORESIST

N-gram baseline top-5

1. STRIP RESIST
2. STRIP PHOTORESIST
3. DEVELOP PHOTORESIST
4. SOFT BAKE
5. SPIN COAT PHOTORESIST



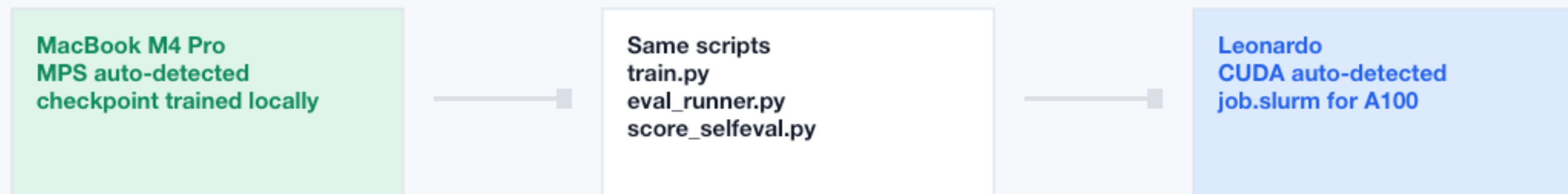
Trained GPT top-5

1. STRIP PHOTORESIST
2. STRIP RESIST
3. STRIP RESIST LEVEL 2
4. FINAL CLEAN
5. OPEN PAD WINDOW

The GPT puts the exact next step at rank 1.

Reproducible on a laptop or Leonardo

The repository contains the checkpoint, loss curve, scorer outputs, and exact commands.



Submission-readiness checklist

- Public GitHub repository
- MIT license
- REPORT.md and requirements.txt
- Official-format CSV IDs
- No official eval sequences used for training

What we would trust in production

Learned process probabilities, explicit rules, and measured OOD behavior.

0.999

Next-step

OOD Top-5

0.263

Completion

self-eval NED

1.00

Anomaly

F1 + rules

15

Families

12 train / 3 test

Kremsians built this as a full benchmark, not just a single CSV: baseline, trained model, OOD protocol, official scorer reports, checkpoint, and demo are all in the repository.

Next engineering step: validator-guided beam search for completion.